PHY 102: Introduction to Physics-2

Tutorial-1

(Vector calculus)

1. **<u>Gradient and divergence:</u>**

   (a). Find the directional derivative of the field $f = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}$ at a point (x,y,z) in the direction of the vector $\vec{r}$ = (x,y,z). Here a, b and c are constants.

   (b). In what case the directional derivative of part (a) is equal to the modulus of the gradient?

2. **<u>Divergence and curl of vector function '$\vec{v}$':</u>**

   (i). $\overrightarrow{v_1} = x^2\hat{\imath} + 3xz^2\hat{\jmath} - 2xz\hat{k}$

   (ii). $\overrightarrow{v_2} = y^2\hat{\imath} + (2xy + z^2)\hat{\jmath} + 2yz\hat{k}$

   (iii). $\overrightarrow{v_3} = \frac{y}{x^2}\hat{\imath} - \frac{1}{x}\hat{\jmath}$

3. **<u>Irrotational vector field:</u>**

   Find a function g(z) such that the vector field "$\vec{v} = 2y\hat{\imath} + \big(2x + g(z)\big)\hat{\jmath} + 3yz^2\hat{k}$ " is irrotational.